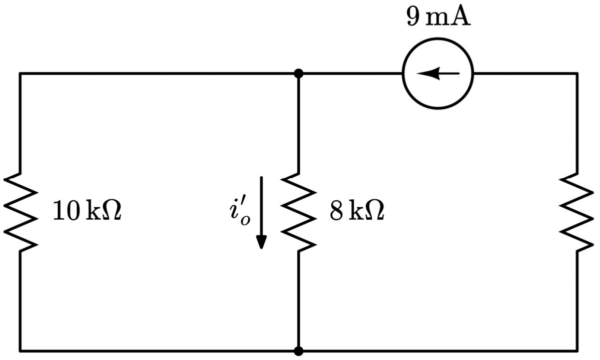
Practice Quiz Three Solution: Superposition (Easy)

For the circuit shown below, use the principle of superposition to find i_o. Clearly redraw the circuit in each state.

Response One: Voltage Source Turned Off (short)



Current Division:

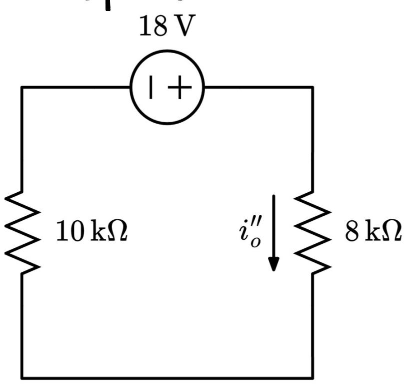
$$i_o' = \underbrace{\frac{10k\Omega \parallel 8k\Omega}{8k\Omega}}_{\text{Unitless Ratio}} \cdot 9mA$$

The 1kΩ resistor is not used for current division since the current through it is known.

$$10k\Omega \parallel 8k\Omega = \frac{40}{9}k\Omega$$

$$i_o' = 5A$$

Response Two: Current Source Turned Off (open)



Ohm's Law:

$$i_o'' = \frac{18\,V}{10k\Omega + 8k\Omega}$$

$$i_o'' = 1mA$$

Combine Responses: $i_o = i_o' + i_o'' = 6mA$

Self-Grading Rubric	
Clearly redraw the circuit correctly when the voltage source is turned off (Voltage source at 0V is shorted)	+2 points
Clearly redraw the circuit correctly when the current source is turned off (Current source at 0A is open)	+2 points
Correct value for i_o' in response one	+2 points
Correct value for i_o'' in response two	+2 points
Correct value for the sum of responses i_o	+2 points